

FIT3155: Week 10 Lab Sheet

(Questions are based on Week 9 topic dealing with B-trees)

Objectives: Develop an understanding of the B-tree data structure and operations.

Conceptual exercises

(Note: In this lab sheet, the terms *keys* and *elements* are used interchangeably. In practice, however, an element consists of a key together with associated payload data.)

1. Write pseudocode for a function `search( $P, x$ )` in a B-tree, where P is any node of a B-tree and x is the key being searched for in the (sub)B-tree rooted at P . The function should first *binary search* the keys stored in node P . If $x \notin P$, the function recursively searches an appropriate subtree of P where it exists. The return value is true/false indicating whether or not x was found in the subtree rooted at P . (A driver function is assumed to initiate the search with P as the root node of the full B-tree.)

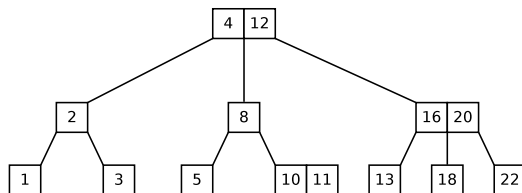
Justify that your search procedure terminates correctly.

2. For a B-tree with minimum degree $t = 3$, insert the following sequence of keys in order:

$\{S, Z, G, Y, B, N, D, E, F, U, I, V, M, X, H\}$.

Draw the state of the B-tree after each insertion.

3. The figure below shows the current state of a B-tree with minimum degree $t = 2$. Delete the keys in the following order: $\{1, 22, 16, 8, 18, 5\}$, and draw the state of the tree after each deletion.



4. For a B-tree $B(t)$ with a minimum degree parameter t containing N elements, derive a **lower bound** on the height h of $B(t)$ as a function of t and N .

5. For a B-tree $B(t)$ with a minimum degree parameter t containing N elements, derive an **upper bound** on the height h of $B(t)$ as a function of t and N .
6. Can a B-tree $B(t)$ with $N \geq 2t$ elements ever reach a state in which **all** nodes are fully occupied (or, conversely, **all** nodes at minimum possible occupancy) through a sequence of insert and delete operations? Justify your answer.
7. Consider the following potential function of a B-tree $B(t)$:

$$\Phi = \sum_{\forall P \in B(t)} ((2t - 1) - \text{nKeys}(P)),$$

where $\text{nKeys}(P)$ denotes the number of keys stored in a node $P \in B(t)$.

- (a) Use the potential function to bound the total number of split operations during n insertions.
- (b) Use the potential function to bound the total number of merge operations during n deletions.
8. Suppose that we insert the keys $\{1, 2, \dots, n\}$ into an initially empty B-tree with minimum degree $t = 2$. Provide tight upper and lower bounds on the number of nodes in the resulting B-tree.
9. What is the worst-case space complexity of a B-tree $B(t)$ storing N elements?
10. Analyze the worst-case time complexities of the search, insertion, and deletion operations, respectively, in a B-tree $B(t)$.

Implementation exercise

(A message of caution: ‘Vibe coding’ with GenAI will hinder your learning of algorithms and data structures. The degree (and this unit) you have enrolled into is a specialist degree (unit). By definition it expects one to demonstrate specialist understanding of topics. This understanding comes from making a concerted effort, via thinking, implementing, debugging etc.)

1. Implement a B-tree (in main memory) data structure supporting the following operations:
 - `insert`
 - `delete`
 - `search`

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  END
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